

ADAM, Oman

WMO: 412640

Lat: 22.4997N

Lon: 57.3700E

Elev: 328

StdP: 97.45

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	13.9	-7.1	2.1	27.9	-4.9	2.6	29.1	12.0	22.7	10.3	21.1	2.6	330	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.1	45.7	21.5	44.6	21.6	43.5	21.6	26.6	35.1	26.1	34.8	25.6	34.5	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.8	29.6	24.1	19.8	29.7	23.4	19.0	29.7	85.8	35.2	83.2	34.8	80.9	34.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.6	8.1	DB	9.3	48.3	1.3	0.9	8.4	48.9	7.6	49.5	6.9	50.0	5.9	50.7	
			WB	4.5	27.1	2.3	0.9	2.8	27.7	1.5	28.3	0.2	28.8	-1.5	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	20.6	22.6	26.5	30.7	34.7	36.4	35.8	34.5	32.7	30.0	25.6	21.9
	DBStd	5.84	1.78	2.47	2.24	2.24	1.80	1.98	2.28	2.38	1.80	1.71	1.68	1.98
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	7	3	2	0	0	0	0	0	0	0	0	0	2
	CDD10.0	7068	328	353	510	621	766	793	799	758	682	620	467	370
	CDD18.3	4033	73	122	252	371	507	543	541	500	432	362	217	113
	CDH23.3	59298	461	1118	3014	5391	8401	9297	9050	7933	6606	5005	2206	818
(14) Wind	WSAvg	3.7	3.1	3.5	3.6	3.9	4.2	4.3	4.3	4.5	3.9	3.1	2.8	2.8
	PrecAvg	117	6	10	23	15	6	8	15	18	5	4	4	3
	PrecMax	372	54	84	257	65	44	38	76	77	17	22	29	19
	PrecMin	27	0	1	0	0	0	0	1	0	0	0	0	0
	PrecStd	78	12	17	52	21	9	11	17	19	5	7	8	5
	DB	29.9	34.0	37.4	41.1	46.1	47.4	46.6	45.4	43.1	40.3	35.4	31.8	
	MCWB	15.6	17.6	17.3	19.3	21.7	21.5	22.1	22.1	21.1	19.3	18.6	17.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	28.3	32.1	36.0	39.7	43.7	46.1	45.3	44.1	41.8	39.0	33.8	29.8	
	2% MCWB	16.2	17.0	17.1	18.9	20.9	21.1	22.1	22.2	21.0	19.2	18.0	17.1	
	5% DB	27.0	30.5	34.8	38.7	42.3	44.8	44.1	43.1	40.8	37.9	32.6	28.7	
	5% MCWB	16.3	17.0	17.3	18.8	20.7	20.8	22.2	22.5	21.1	19.1	18.3	17.2	
	10% DB	25.9	29.0	33.5	37.6	41.1	43.5	43.0	41.9	39.7	36.5	31.4	27.5	
	10% MCWB	16.0	16.7	17.2	18.6	20.3	20.9	22.2	22.7	21.2	18.8	18.1	17.2	
	WB	20.0	21.6	23.2	24.7	26.6	27.6	27.1	26.2	24.0	23.6	21.3		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	24.6	26.9	27.6	28.9	33.2	36.0	37.8	36.6	34.7	31.6	28.9	24.7	
	2% WB	18.9	20.5	22.4	23.3	25.8	26.7	26.2	26.2	25.2	23.0	21.8	20.0	
	2% MCDB	23.6	25.6	27.4	28.7	32.0	33.7	37.1	36.4	34.0	30.2	26.6	24.4	
	5% WB	18.2	19.5	21.4	22.5	25.2	26.1	25.7	25.6	24.5	22.3	20.9	19.3	
	5% MCDB	23.5	25.6	27.5	29.4	32.0	33.3	36.2	35.7	33.4	29.7	26.1	24.6	
	10% WB	17.4	18.6	20.2	21.6	24.4	25.5	25.2	25.0	23.9	21.6	20.1	18.7	
	10% MCDB	23.0	25.4	27.3	30.4	32.4	33.4	35.2	34.8	32.7	29.7	26.4	24.5	
(35) Mean Daily Temperature Range	MDBR	11.9	13.0	13.9	13.4	13.5	14.1	14.2	14.6	14.1	13.7	12.3	11.9	
	5% DB	12.7	14.5	15.4	14.5	14.7	15.4	15.0	15.2	14.8	14.7	13.4	12.9	
	MCWBR	4.8	5.2	5.0	4.8	4.6	4.7	3.7	3.0	3.5	4.3	4.7	4.8	
	5% WB	10.6	11.8	11.6	12.7	13.8	14.1	14.5	14.6	14.3	12.9	11.6	10.8	
	MCWBR	4.4	4.7	4.3	4.5	4.9	4.0	3.2	3.1	3.2	4.3	4.8	4.3	
(40) Clear-Sky Solar Irradiance	taub	0.391	0.435	0.495	0.558	0.603	0.688	0.774	0.727	0.603	0.464	0.417	0.389	
	taud	2.203	2.046	1.872	1.761	1.676	1.550	1.453	1.518	1.704	1.989	2.139	2.202	
	Ebn at Noon	865	848	813	769	730	665	610	639	714	808	830	851	
	Edh at Noon	131	163	203	231	250	281	310	290	237	170	138	127	
(44) All-Sky Solar Radiation	RadAvg	4.83	5.62	6.47	7.13	7.65	7.39	6.78	6.79	6.51	5.98	5.03	4.58	
	RadStd	0.17	0.20	0.31	0.32	0.12	0.27	0.29	0.16	0.11	0.15	0.10	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.23	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96	+98

Nomenclature: See separate page